Chapter 1

Explainability in Reinforcement Learning

Today, the majority of artificial intelligence agents with which humans interact are deployed in controlled or low-risk environments. For example, agents capable of playing games such as Go operate in settings where potential errors remain limited in terms of consequences. Even more autonomous systems, such as agents based on large language models, remain constrained by human supervision mechanisms and are often restricted to tasks such as code generation or information retrieval. Safety-critical applications integrating machine learning components are still rare. Introducing autonomous agents in such environments, requires the establishment of strong guarantees. These requirements include in particular:

- validating the system's functioning before deployment,
- identifying the origin of a malfunction in case of failure,
- and certifying that an observed failure will not occur again.

The main challenge lies in the fact that the performing agents currently available rely on both reinforcement learning and deep learning. As a result, the mechanisms underlying their decisions cannot be understood directly from the learned model. This lack of explainability makes their use difficult in safety-critical contexts. To address this limitation, a research field has emerged: Explainable Reinforcement Learning (XRL), a subfield of Explainable Artificial Intelligence (XAI). The objective of XRL is to develop explainability methods that make it possible to obtain explanations of reinforcement learning agents.

This chapter presents the main existing XRL methods. We first define what is meant by an explanation in the context of XAI. We then propose a taxonomy for organizing existing XRL methods. This taxonomy serves as the basis for the state-of-the-art review presented in the remainder of the chapter.

1.1 Explanation in Artificial Intelligence

In this section, we define what an explanation is in the context of XAI and how it can be obtained. We first introduce the concept of explanation in a general.

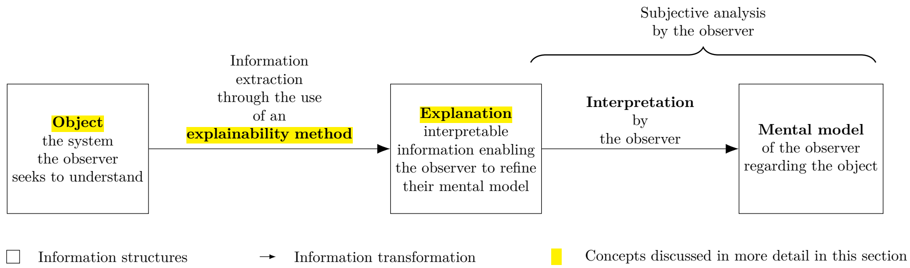


Figure 1.1: Explanation process in explainable artificial intelligence.

We then describe how this concept applies to XAI. Finally, we introduce the concept of explainability methods and describe the properties that characterize explanations in the context of XAI.

1.1.1 Explanation

Establishing what is meant by an explanation is challenging due to the fundamentally multidisciplinary nature of the concept. When we refer to explanation, we are inherently referring to an explanation provided to a human. As soon as humans are involved, the problem is no longer purely computational or mathematical. It involves several fields of research, such as philosophy, psychology, and sociology. So far, these disciplines have not converged on a unified definition of explanation that can be directly applied to XAI.

For this reason, we propose the following definition of explanation. An explanation involves two entities: the object (Definition 1.1.1) being explained and the receiver of the explanation. The receiver is a human who aims to understand how the object operates and will be referred to as the observer throughout this manuscript (Definition 1.1.2). The observer seeks to understand the object and, in doing so, builds a mental model of its functioning. An explanation is therefore what allows the observer to refine and correct their mental model of how the object operates (Definition 1.1.4).

For an observer to understand an explanation, the information it contains must be interpretable (Definition 1.1.3). Information is considered interpretable when it can be directly understood by a human without requiring any additional transformation process.

Definition 1.1.1: Object

The object of an explanation is the system is to be understood by an observer.

Definition 1.1.2: Observer

The observer is the human who aims to understand a object. The observer possesses a mental model of the object being explained. This mental model is an internal representation of how the observer believes the system operates. This representation may be incomplete or inaccurate.

Definition 1.1.3: Interpretable

Information is interpretable when it can be directly understood by an observer without requiring any additional transformation process.

Definition 1.1.4: Explanation

An explanation is any interpretable information that enables an observer to refine or correct their mental model of the object of the explanation.

This definition of explanation provides the foundation for the approach to explainability adopted throughout this manuscript. Its generality allow it to encompass the different explainability methods considered in the following sections.

Moreover, it avoids a binary interpretation of explanation, where a system is considered either explainable or not explainable. From our perspective, the notion of non-explainable system is not meaningfule. Considering a system as impossible to explain would therefore contradict the general objective of scientific understanding.

This quantitative perspective allows us to express an intuitive idea: not all explanations provide the same level of value. The effectiveness of an explanation depends on its ability to improve the observer's understanding of the system. For example, describing a marginal aspect of a system is different from correcting an observer's misunderstanding of a key mechanism.

Figure 1.1 summarizes this whole explanation process, from the object of the explanation to the mental model refined by the observer.

1.1.2 Explainable Artificial Intelligence

Now that the general framework of explanation has been established, we can examine how this concept applies in the context of artificial intelligence. In the context of XAI, the object of explanation is an artificial intelligence system. More precisely, XAI emerged as a response to the success of deep learning models. These models achieve high performance through connectionist architectures based on deep neural networks. In these architectures, information is represented in a distributed manner across the network through patterns of activation and connections between computational units. Unlike symbolic approaches, individual components of neural networks cannot be directly associated with human-defined concepts. From this starting point, two types of approaches emerge:

- **Substitutive explainability:** In this approach, the underlying assumption is that there is nothing to explain within a neural network. There is no semantic meaning associated with each computational component, and therefore nothing to interpret. This corresponds to the so-called "black box" view of neural networks. Under this assumption, XAI consists of replacing as much of the deep learning system as possible with alternative AI methods.

- **Intrinsic explainability:** In this approach, it is argued that neural networks inherently contain mechanisms that can be explained. The limitation is considered to lie not in the networks themselves, but in the absence of adequate tools to analyze their functioning.

In this manuscript, we present methods stemming from both approaches to explainability. This manuscript considers methods from both approaches in order to provide a comprehensive overview of the XAI field. Throughout this work, we adopt the intrinsic explainability perspective as the underlying viewpoint. This perspective influences how we define explainability and how we organize the analysis of existing methods.

1.1.3 Explainability Method

Now that the general framework of XAI has been established, we need to further detail the properties of an explanation in this context. In our framework, an explanation in the field of XAI can be characterized by three components:

- **Source:** refers to the origin of the information used by the explainability method. The explainability method transforms the raw information produced by the AI system into an explanation that can be understood by an observer.
- **Form:** refers to the representation through which the explanation is communicated. The form defines how the information is presented to the observer and must allow human interpretation. An explanation can take different forms, such as text, images, computer code, mathematical equations.
- **Subject:** refers to the aspect of the AI system that is being explained. In XAI, this distinction is commonly associated with the difference between local and global explanations. A local explanation describes why an AI system produced a specific output in a given situation. A global explanation aims to describe the general functioning of the AI system.

An explainability method can therefore be viewed as a transformation process that maps a source of information into an explanation characterized by a specific form and subject. Current research in XAI focuses on developing new methods capable of performing this transformation in order to overcome the lack of interpretability associated with deep learning models.

1.2 Taxonomy of XRL Methods

Now that the concept of XAI has been defined, we turn to its application in reinforcement learning, known as XRL. As in XAI, explanations in XRL are produced by explainability methods and can be described in terms of their subject, source, and form. This section reviews the possible variations of these three aspects in the specific context of reinforcement learning.

1.2.1 Subject

The first question to address is the subject of explanation in the context of reinforcement learning. To answer this question, it is necessary to identify the different components involved in a reinforcement learning system and determine which of them constitutes the object of explanation. A reinforcement learning system can be described through two main components: the environment and the policy.

The environment defines the dynamics in which the agent evolves, including the possible states, observations, actions, and transitions. Environments are simulations designed by humans, and their dynamics are explicitly defined. Therefore, it is assumed that the environment is known to the observer and does not require explanation.

The main object of interest is therefore the agent's policy. The policy represents the decision-making mechanism of the agent. It is a function that maps observations to actions and is learned through interaction with the environment in order to maximize the expected cumulative reward. The policy is the component that encodes the learning process and that XRL aims to explain.

Explaining a policy introduces additional challenges compared with classical XAI settings. In traditional XAI problems, the objective is to explain an isolated decision, such as determining which features of an image led to a classification result. In contrast, reinforcement learning introduces a temporal dimension into the decision-making process. An action is not only determined by the current observation but also influences future states and rewards. Therefore, understanding the behavior of an agent may require considering a sequence of interactions between the agent and its environment.

This temporal dimension leads to a distinction between two objectives of explanation in XRL:

- **Decision explanations:** focus on identifying the elements of the current observation that influenced the action selected by the policy. This type of explanation is close to classical XAI approaches, as it analyzes the relationship between inputs and outputs while abstracting away the sequential nature of the decision process.

- **Strategy explanations:** aim to characterize the agent's behavior over time. They focus on sequences of decisions and interactions with the environment in order to understand how the agent achieves its long-term objective. This type of explanation is specific to reinforcement learning, as it addresses the temporal and goal-oriented nature of the learning process.

1.2.2 Source

We identify three sources of explanation in reinforcement learning, all obtained during or after the learning process and containing information about the agent:

- **Policies:** policies are functions that generate a distribution over the action space for a given observation. The objective of reinforcement learning is to learn a policy that maximizes the expected cumulative future rewards. During the learning phase, the policy is iteratively improved through interactions with the environment.

- **Trajectories:** trajectories represent sequences of successive interactions between the agent and the environment. At each step, the agent selects an action based on its observation, which modifies the environment and produces a reward as well as a new observation. This process continues until the end of the trajectory.

- **Value functions:** value functions provide predictions about the future outcome of a trajectory from a given observation. The most commonly used value function is the critic function, which estimates the expected sum of future rewards. This prediction guides the learning process, as the agent updates its policy to maximize the estimated future return. However, other types of value functions can also be used as sources of explainability.

1.2.3 Form

We identify six forms of explanation in the RL setting:

- **Policy Model:** an interpretable model of the agent's policy function.
- **Saliency Map:** a weighting of the input features representing their influence on the output of the policy function.
- **Counterfactual:** a representation of a modified observation obtained by perturbing the original observation in order to produce a different action from the agent.
- **Agent Trajectory:** a representation of a sequence of successive interactions between the agent and the environment.
- **Trajectory Prediction:** a prediction about the future evolution of the trajectory produced by a value function.
- **Interaction Model:** an interpretable model of the interactions between the agent and the environment.

Among the three components characterizing an explanation, the form is the one that varies most significantly across explainability methods. While subjects and sources tend to be relatively stable in the XRL literature, the diversity in how explanations are presented is considerably greater. The list above is by no means definitive and may evolve as new approaches emerge. For this reason, the following section is organized according to the form of explanations and presents existing explainability methods based on this criterion.

1.3 State-of-the-Art Review of XRL Methods

The purpose of this section is to detail the functioning of existing explainability methods (see Table 1.1 and Figure 1.2). They are presented according to the form of explanation they produce, following the taxonomy introduced in Section 1.2. Each subsection introduces one of the six forms of explanation identified in Section 1.2.3, as well as the methods associated with it.

Explanation Target	Explanation Form	Explainability Method	Source of Explanation		
			Policy	Trajectories	Value functions
Decision	Policy Model	*Interpretable policy learning*	✓		
		Policy approximation via interpretable model	✓		
	Saliency Map	*Observation perturbation*	✓		
		Observation gradient analysis	✓		
		Attention-based architecture	✓		
	Counterfactual	*Latent space usage for counterfactual generation*	✓		
Strategy	Trajectory Prediction	*Value functions as observations*			✓
		Critic enrichment			✓
	Interaction Model	*Bayesian network learning*		✓	
		Latent space usage for MDP generation	✓		
	Agent Trajectory	*Hierarchical decomposition*	✓		
		Extraction via critic analysis		✓	✓
		Extraction via multi-critic comparisons		✓	✓

Table 1.1: Taxonomy of explainability methods in reinforcement learning.

1.3.1 Policy Model

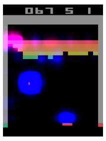
There exists a category of models that can be understood through the simple analysis of their representation; such models are referred to as interpretable [Bellucci et al., 2021]. These models are used to represent the policy function. The main models considered interpretable include decision trees [Topin et al., 2021], algorithms [Trivedi et al., 2022], formal logic [Payani and Fekri, 2019], and linear models [Garreau and von Luxburg, 2020]. The intrinsic interpretability of these models can be leveraged to provide explainability in the context of reinforcement learning (RL).

This directly interpretable representation allows an observer to extract information. Explanations derived from an interpretable model primarily provide insights into the decision-making process, since what is modeled is the mapping between an observation and an action. If the model is sufficiently simple or well structured, the observer can form a mental representation of a sequence of decisions across the environment and thus obtain information about the agent's strategy.

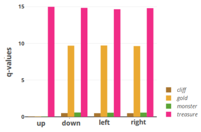
To obtain an interpretable policy model, two explainability methods are available: *interpretable policy learning* and *policy approximation via interpretable model*.

Interpretable Policy Learning

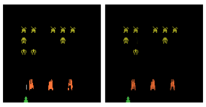
This method aims at learning a policy represented by an **interpretable** model [Topin et al., 2021, Trivedi et al., 2022]. It is the most direct approach to addressing the challenges posed by explainable reinforcement learning (XRL). This type of training requires modifying the MDP in order to adapt it to the search space and to the type of policy representation. Moreover, this search space often needs to be reduced to make learning feasible. Such a **reduction** requires the integration of **expert knowledge** into the learning process. With this method, the *interpretable policy function* itself is used to provide explanations to the observer.



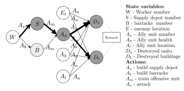
(a) Perturbation-based heatmap for the Atari game "Breakout" [Greydanus et al., 2018]. Highlighted areas correspond to the most influential regions for the policy function's decision.



(b) Prediction of sub-rewards as a function of the action selected by the agent for a given state [Juozapaitis et al., 2019]. Depending on the action chosen by the agent, one can determine which sub-reward it aims to maximize.



(c) Counterfactual explanation of the agent's behavior for the game *Space Invaders* [Olson et al., 2021]. In the left image, the observation leads to the action "LEFT", whereas in the right image, the action selected by the agent is "RIGHT_FIRE".



(d) Agent–environment interaction model for the video game *StarCraft II* [Madumal et al., 2020]. Nodes represent random variables, and edges represent causal relationships between them.

Figure 1.2: Overview of XRL explanation methods. **(a)** Saliency map, **(b)** decomposed Q-values, **(c)** counterfactual states, **(d)** causal graph.

Policy Approximation via Interpretable Model

Using an interpretable model, it is possible to **approximate** a non-interpretable policy [Sieusahai and Guzdial, 2021]. In this setting, the problem is cast as a **supervised learning** task, where the non-interpretable policy is used to label each observation with the action it selects. Using this dataset, supervised learning is then performed with the interpretable model. The resulting *interpretable model* is subsequently used as an explanation for the observer. If this approximation is sufficiently faithful, it can be deployed in the environment in place of the original policy, thereby enabling a high degree of control over the agent.

1.3.2 Saliency Map

Heatmaps make it possible to identify which parts of the input vector are decisive in the selection of the output (as illustrated in Figure 1.2a). When applied to the reinforcement learning setting, they highlight the regions of the observation space that are critical in the action selection process. By nature, this form of explanation provides information about the policy's decision-making process. The intuitive nature of this representation allows it to be used by a broad audience. Three explainability methods can be used to obtain this type of explanation: *observation perturbation*, *observation gradient analysis*, and *attention-based architectures*.

Observation Perturbation

Perturbation analysis consists in applying various **modifications** to the observation in order to study their **impact** on the output of the **policy function** [Greydanus et al., 2018]. The greater the variation in the output caused by a perturbation applied to a given component of the input vector, the higher the **weight** assigned to that component. The magnitude of the perturbation is measured by computing the distance between the original output and the output obtained from the perturbed input. Using this set of weights, a *heatmap* is constructed, enabling the observer to visualize which parts of the observation vector are most sensitive in the action-selection process.

Observation Gradient Analysis

Observation **gradient** analysis refers to a set of methods that require a differentiable policy function in order to extract information from it [Simonyan et al., 2014, Weitkamp et al., 2019, Huber et al., 2019]. To this end, the gradient of the policy function is computed with respect to all **components of the input vector**. For each input component, the resulting gradient provides a weighting that represents its importance in the output of the policy function. Based on these weightings, a *heatmap* is constructed to allow the observer to visually identify the parts of the input vector that are the most influential in the policy function's output.

Attention-Based Architecture

Attention blocks constitute a neural network architecture that makes it possible to weight the relative importance of one part of a vector with respect to the other parts of the same vector [Vaswani et al., 2023]. These weights are computed as a function of the vector itself and are learned by the neural network during training. For a given observation, each component of the observation vector is assigned a scalar value. It can therefore be assumed that the components associated with the highest weights are the most influential in determining the output [Mott et al., 2019, Itaya et al., 2021]. This attention-based *heatmap* is then provided to the observer as the explanation.

1.3.3 Counterfactual

In the context of reinforcement learning, counterfactual explanations are used to generate new observations (see Figure 1.2c). This new observation, distinct from the original one, leads the agent to adopt an alternative behavior. A counterfactual explanation aims to explain the agent's decision-making process. Modifying the observation informs the observer about the adjustments required for the agent to adopt a different behavior. The method available to generate counterfactuals is the *use of the latent space for counterfactual generation*.

Latent Space Usage for Counterfactual Generation

In this method, **generative models** are used [Bond-Taylor et al., 2021] to exploit the latent representation. This type of model learns to generate new data by imitating the training data distribution. In this context, generative models aim to produce alternative observations that are likely to alter the agent's actions [Olson et al., 2021].

The main challenge of this approach lies in the fact that, to serve as explanations, counterfactuals must be both **close** and **feasible**. A close counterfactual refers to one that does not differ significantly from the original observation. A feasible counterfactual corresponds to an observation that is meaningful within the considered environment.

1.3.4 Agent Trajectory

Trajectories represent a sequence of interactions between the agent and its environment. By analyzing these trajectories, the observer can infer the underlying dynamics governing the agent's evolution within its environment.

Post-training trajectories are systematically used to obtain explanations of the agent's strategy, even outside the scope of explainable reinforcement learning (XRL). This simple approach can be enhanced using methods such as *hierarchical decomposition* and *extraction via critic analysis*.

Training trajectories, on the other hand, can be used to explain the agent's learning process, through the *extraction via multi-critic comparisons* method.

Hierarchical Decomposition

Hierarchical agents are composed of a set of **sub-policies** specialized for specific tasks within subsets of the environment's states. This method distributes the optimization of the reward function across multiple models, significantly simplifying the learning process. Moreover, this **decomposition** of the policy function can also provide explainability in the agent's strategy [Beyret et al., 2019]. If a sub-policy is selected, it indicates that the agent will execute a specialized behavior. This behavioral specialization provides insight into the strategy the agent employs to maximize the reward function. The agent's *trajectory*, together with the *sub-policy* selected for each action, is used as interpretable information.

Extraction via Critic Analysis

In this method, the predictions of the critic are used to determine which **trajectories to present to the observer** [Sequeira and Gervasio, 2020]. A set of states is selected from multiple trajectories. For a given state, all possible actions are examined. The state is assigned a value equal to the difference between the critic's prediction for the lowest-valued action and the prediction for the highest-valued action. Trajectories containing the states with the highest values according to this procedure are then chosen to be presented to the observer. These trajectories are considered **critical moments** of the agent and are therefore relevant for analysis. For this method, the *selected trajectories* represent the interpretable information provided to the observer.

Extraction via Multi-Critic Comparisons

In this approach, the main objective is to **weight the training interactions** between the agent and the environment according to their influence on the learning process. To achieve this, multiple critic functions are created using *off-policy* learning. Off-policy learning is a reinforcement learning process that allows a policy to learn from data generated by a different policy [Gottesman et al., 2020]. Each critic function is trained on a subset of the dataset.

A reference critic function is trained using the entire dataset. Then, for each agent-environment interaction, a critic function is retrained while specifically excluding that interaction. The **prediction disparity** between the retrained critic and the reference critic represents the impact of removing that interaction on learning. Through this procedure, a value is assigned to each interaction based on this difference: the larger the disparity, the higher the value assigned to the interaction.

A higher value indicates that the interaction is considered more important in the learning process. These *values* over the *interactions* serve as explanations for the observer, providing a means to assess the significance of each agent-environment interaction within the learning process.

1.3.5 Trajectory Prediction

Making predictions about the future of a trajectory allows one to form a representation of what the system expects, based on its training experience (see Figure 1.2b). These predictions are produced using value functions. Such functions provide insights into the future evolution of the agent's trajectory. By their nature, this form of explanation informs the observer about the strategy adopted by the agent.

The main challenge of this type of approach lies in ensuring that the values predicted by these functions are truly decisive in the actions selected by the agent. To this end, the following methods are employed: *value functions as observations* and *critic enrichment*.

Value Functions as Observations

The objective is to train **interpretable value functions** to be used as observations for the agent. If the features predicted by these value functions are **meaningful** to the observer, they can determine which aspects the agent is attempting to maximize through its actions [Lin et al., 2021].

Critic Enrichment

In order to obtain interpretable information from the predictions of the **critic**, it is possible to **decompose** the critic into multiple **sub-functions** [Juozapaitis et al., 2019]. The reward function is thus decomposed into multiple sub-reward functions, each representing a **sub-objective** of the task to be accomplished. Similarly, for each of these sub-reward functions, corresponding sub-critic functions are defined, along with a function that combines their outputs. During training, the agent selects its actions so as to maximize this combination function of the sub-critics. Using this approach, for each observation and action, a prediction is obtained regarding the sub-objectives the agent is attempting to maximize. This *prediction* is then used as the explanation for the observer.

1.3.6 Interaction Model

Modeling the interactions between an agent and its environment makes it possible to understand how the agent's decisions are made and what impact they have on the environment (see Figure 1.2d). By incorporating these interactions into an explanatory model, one can better grasp the complex relationships between the agent and its environment, thereby identifying the determining factors in the decision-making process. By explicitly handling the agent–environment interaction, this type of explanation aims to explain the agent's strategy. The methods used to obtain explanations in the form of agent–environment interactions are: *Bayesian network learning* and *latent space usage for MDP generation*.

Bayesian Network Learning

The evolution of an agent within an environment can be observed as a set of **random variables** connected through a **Bayesian network** [Madumal et al., 2020]. Random variables are defined to represent both external aspects (the environment) and internal aspects of the agent (the actions). By analyzing trajectories, it becomes possible to infer causal relationships among these random variables. At the end of this process, a weighted causal graph is obtained, which serves as an explanation for the observer. This *Bayesian network*, representing the interactions between the agent and the environment, is then used as the explanation.

Latent Space Usage for MDP Generation

Each layer of a neural network can be viewed as a projection from one space to another [Zahavy et al., 2017]. As information propagates through successive layers of a neural network, it becomes increasingly abstract. It can therefore be assumed that two vectors that are close in this projected space are also close in terms of the agent's perception. Based on this observation, it is possible to **redefine an MDP** by leveraging the **abstract representations** produced by the final layers of a neural network. Once the new MDP has been constructed, a model of the interaction between the agent and its environment is obtained. This model, represented as a *graph*, is then used as the explanation.

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